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Applied AI researcher and engineer building production-grade AI systems across LLMs, retrieval, agentic workflows, and evaluation infrastructure: from research to reliable real-world deployment.

Professional Experience

AI Lead
Oct. 2025 – Present

ErisAI

- Architect and develop **Synapse**, a research-driven AI infrastructure layer for multi-agent workflows, integrating hybrid retrieval, multi-turn memory, MCP-based tool orchestration, intelligent model routing, and evaluation and observability protocols using Ragas and LangFuse to catch traceability and knowledge-access failures. (Synapse)
- Deployed **Synapse**, incorporating a LoRA fine-tuned model via Unsloth, across multiple enterprise AI applications including legal and investigative platforms, processing over 30M+ private documents with grounded retrieval and traceable responses.
- Built **Ownify AI**, a multi-tenant white-label platform on top of Synapse with structural tenant isolation, per-tenant billing and usage metering, and workspace branding for organizations running their own AI product. (Product)

AI Researcher
Sep. 2024 – Oct. 2025

VOLV AI

- Researched and developed a 3D virtual try-on SDK using neural rendering, GANs, and diffusion models; the system was adopted by fashion brands and increased user engagement by 40%.
- Developed controllable 2D and 3D digital human systems with pose estimation and generative modeling, supporting personalized experiences and reducing product return rates by 30%.
- Prototyped multimodal LLM systems integrating language, vision, and audio for cross-modal reasoning and instruction-following in customer support and interactive applications.

Research AI Engineer
Jan. 2023 – Dec. 2023

BHuman AI

- Built scalable generative video pipelines using neural reenactment, facial motion transfer, and voice cloning, reducing production costs by 40% for a platform serving 90K+ users and enterprise media clients. (Use cases)
- Researched and deployed audio-driven lip-sync models using GANs and super-resolution, improving visual fidelity and establishing a core product capability and primary revenue stream.
- Integrated LLMs with persona-driven avatars and developed retrieval-based conversational systems for enterprise clients, combining private knowledge, persona conditioning, and response evaluation for real-world deployments.

Intern Machine Learning Engineer
Aug. 2021 – Nov. 2021

NAECO Blue GmbH [*Profile*]

- Developed machine learning models to improve the spatial and temporal resolution of weather data for solar and wind energy forecasting.
- Built a data pipeline that reduced R&D time by nearly 50% and supported forecasting workflows later adopted by 5+ European meteorological agencies.

Selected Projects

Synapse (2026). Research-driven AI system integrating retrieval, memory, routing, evaluation, and runtime controls for reliable model behavior.

Ownify AI (2026). Multi-tenant white-label AI platform built on Synapse, with structural tenant isolation, per-tenant billing, and operational tooling for organizations running their own AI product.

PatchProof (2026). Framework for evidence-grounded patch verification using execution traces, behavioral validation, and correctness analysis beyond unit tests.

DarwinPatch (2026). Verifier-grounded coding-agent repair framework combining bounded evidence, route-aware search, and regression gates, lifting solve rate from 0.6 to 0.9.

GateGRPO (2026). Verifier-grounded reinforcement learning (GRPO) for code repair, using the hard-gate verifier itself as the reward signal in place of a learned reward model.

GateDPO (2026). Verifier-ordered DPO for code repair without a learned reward model, showing task-corpus diversity, not the preference algorithm, controls out-of-distribution generalization.

GateRSI (2026). Recursive self-improvement (RSI) for knowledge agents, routing memory, curriculum, and GateDPO weight updates through one verifier, lifting solve rate up to 3.3x with held-out and external-benchmark transfer.

WarmLead (2026). Cache-liveness scheduler for lead-plus-sidekick LLM agent harnesses, with a cost simulator and live API verification to reduce idle-period token spend.

FusionLine (2026). TypeScript benchmark harness for MCP tool-selection routing, identifying a fusion-boundary rule for reranker combination and cost-verifying results against live provider APIs.

DeciForm (2026). Structured extraction library compiling a Pydantic schema into batched decision-model questions, with calibrated per-field confidence and a generative fallback for only the low-confidence fields.

Callona (2026). Self-hosted speech-to-speech AI receptionist answering business calls in under two seconds, using forced-choice tool routing and deterministic coordinators to keep an 8B model reliable.

Skills

- **Programming**: Python, TypeScript, C++, Bash, MATLAB, SQL/NoSQL (PostgreSQL, MySQL, Redis)
- **LLM Systems**: RAG, Multi-Agent Systems, Autonomous Agent Orchestration, MCP (Model Context Protocol), Function Calling & Tool Use, Conversation Memory, Prompt Engineering (few-shot, structured output, system-prompt design), Prompt Caching & Context Window Optimization, Human-in-the-Loop Workflows, LLM Guardrails & Safe Action Design, Inference Optimization (batching, latency and cost), vLLM, llama.cpp, Ollama
- **Retrieval & Grounding**: Hybrid Retrieval, Vector Search (Qdrant, FAISS, pgvector), Cross-Encoder Reranking, Hallucination Detection (NLI Verification), Structure-Aware Document Chunking, Document Intelligence (OCR, PDF and scanned-document extraction, Docling, RapidOCR, stale-layer detection & text-layer rescue), Evidence Provenance & Traceability, Zero-Shot Information Extraction, Ragas, LangFuse
- **Voice & Real-Time Systems**: LiveKit Agents SDK, WebRTC, SIP Trunking, Speech-to-Speech Models, Silero VAD, Neural Vocoders, Asyncio (Real-Time I/O)
- **Frameworks**: LangChain, LangGraph, Haystack, FastAPI, Pydantic, Unsloth, TRL, OpenAI API, Anthropic API, Gemini API
- **ML Fundamentals**: PyTorch, Transformers, TensorFlow, Scikit-learn, NumPy, Pandas, Deep Learning, Statistical Modeling, Feature Engineering, large-scale data processing, multimodal learning, LoRA/PEFT fine-tuning, Quantization, SFT, Reinforcement Learning (RL, DPO, GRPO), Synthetic Data Generation, Classification, Ranking, model and LLM evaluation metrics, benchmark task design (MLE-Bench-style), experiment design
- **Infrastructure**: Docker, Kubernetes, Git, Linux, CI/CD (GitHub Actions), MLOps (model versioning, monitoring, logging, rollback), Systemd, Weights & Biases, Prometheus, Grafana, Async Processing & Caching (Redis), AWS (Bedrock, AgentCore, Lambda, S3, API Gateway, IAM, CloudWatch, ECS, EKS, RDS, DynamoDB), Azure, GCP
- **Computer Vision / 3D**: OpenCV, PyTorch3D, MediaPipe, ONNX

Publications

Taneem, U. J., & Ayesha, N. (2024). Beyond CNNs: Encoded Context for Image Inpainting with LSTMs and Pixel CNNs. International Journal of Innovations in Science & Technology, (IJIST) 6(5), 165–179, Special Issue ICTIS 2024.

Taneem, U. J., & Zakira, I. (2022). HTML Code Generation from Images with Deep Neural Networks. Journal of Engineering and Applied Sciences (JEAS), UET Peshawar. **(Young Undergraduate Researcher Award)**

Education

University of Engineering and Technology Peshawar, Pakistan

- Bachelor of Science in Computer Science — Sep. 2018 – Sep. 2022
- CGPA: 3.58/4.0
- Young Undergraduate Researcher Award